Approval System: how a request flows, step by step

A walkthrough built from passing end-to-end tests · report date 2026-10-21

> **103 end-to-end tests, all passing:** 44 rejection-and-recovery cases, 48 business-workflow tests and 11 admin-security tests. Every screenshot below was taken by one of these tests in a real browser.

Results at a glance

Area	What it covers	Result	Last run	Run on the final code?
Rejection and recovery	44 cases: reject at any stage, fix, resubmit, finish	**44 / 44 passed**	2026-10-21	Yes
Who may start Final (T08E, T10C)	The parent department's manager is denied; the requester's own unit starts Final	**2 / 2 passed**	2026-10-21	Yes
Business workflow suite (T01–T20)	End-to-end journeys, rules and safeguards	**46 / 46 passed**	2026-10-06	No: before the Oct 19–21 review fixes
Admin security	Who can reach and change the admin area	**11 / 11 passed**	2026-09-21	No
Code checks on the merged main	Type check, 78 tool tests, 3,364 regression tests, production build, upgrade rehearsal	**All passed** (3,362 passed, 2 skipped by design, 0 failed)	2026-10-21	Yes

"Run on the final code: Yes" means no product code changed after that run. **"No"** means the test passed on an earlier version and has not been re-run since; its date is shown so nothing looks newer than it is.

The flow in one picture

	Stage	Who acts	Approval chain (default)	Result
1	**Request**	The requester submits a request	Supervisor → Section manager → Department manager	The request goes to Engineering
2	**Solution**	Engineering submits a solution	Engineering's Supervisor → Engineering's Section manager → Department manager	Solution approved; back to the requester
3	**Final**	The requester (or someone in the requester's unit) starts Final approval	Section manager → Department manager	**Completed**

At every approval step the approver can **approve**, or **reject with a reason**. A rejection sends the work back to the right person to fix and resubmit, and the approvals start again. Each stage below shows the normal path first, then what happens after a rejection.

Stage 1: the Request

1.1 The requester submits a request

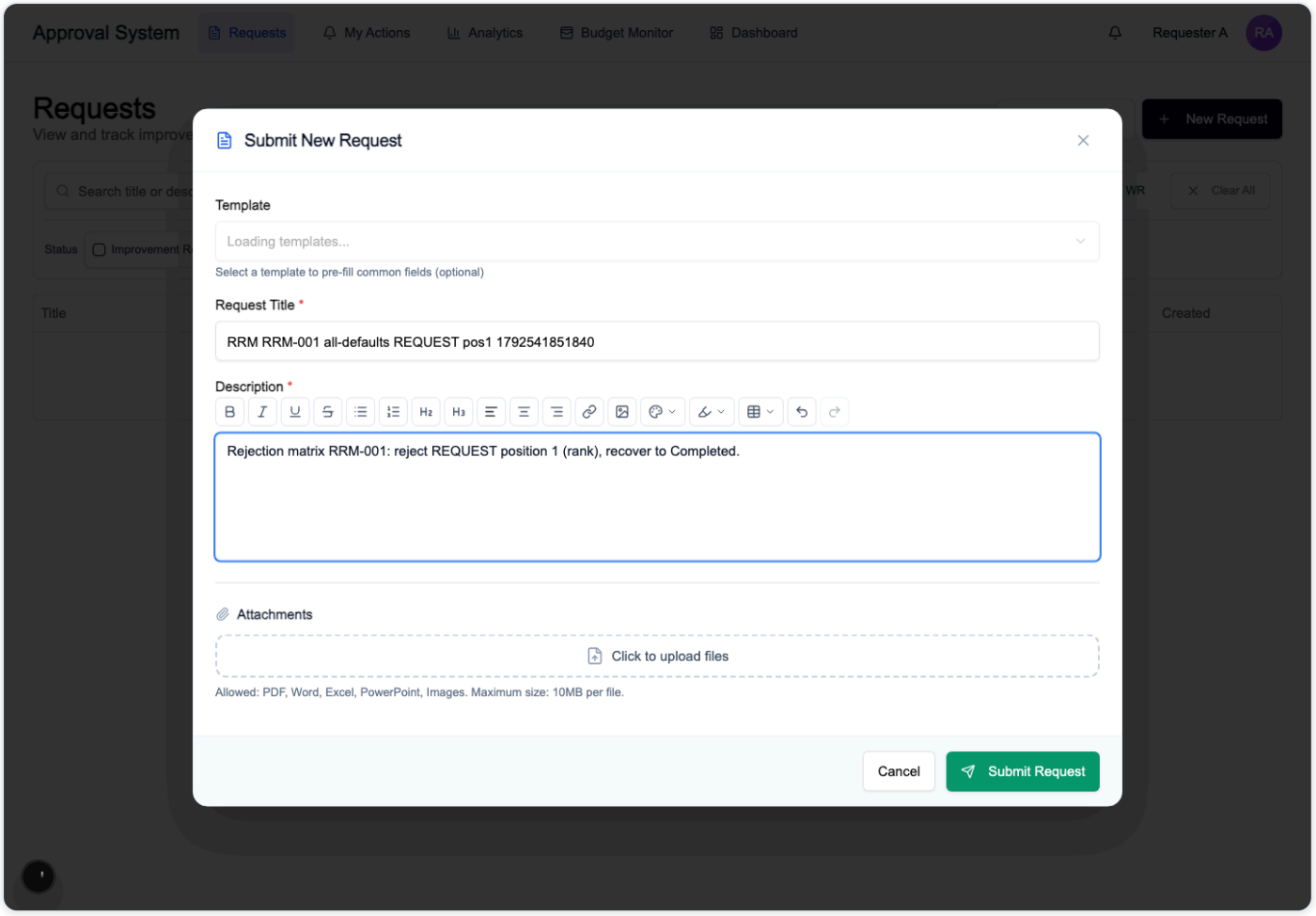


Figure 1. Submit New Request: a title, a description and optional attachments.

The requester chooses **New Request**, writes a title and a description (attachments are optional) and submits. The first approver is notified in the app.

1.2 Approvers decide, one after another

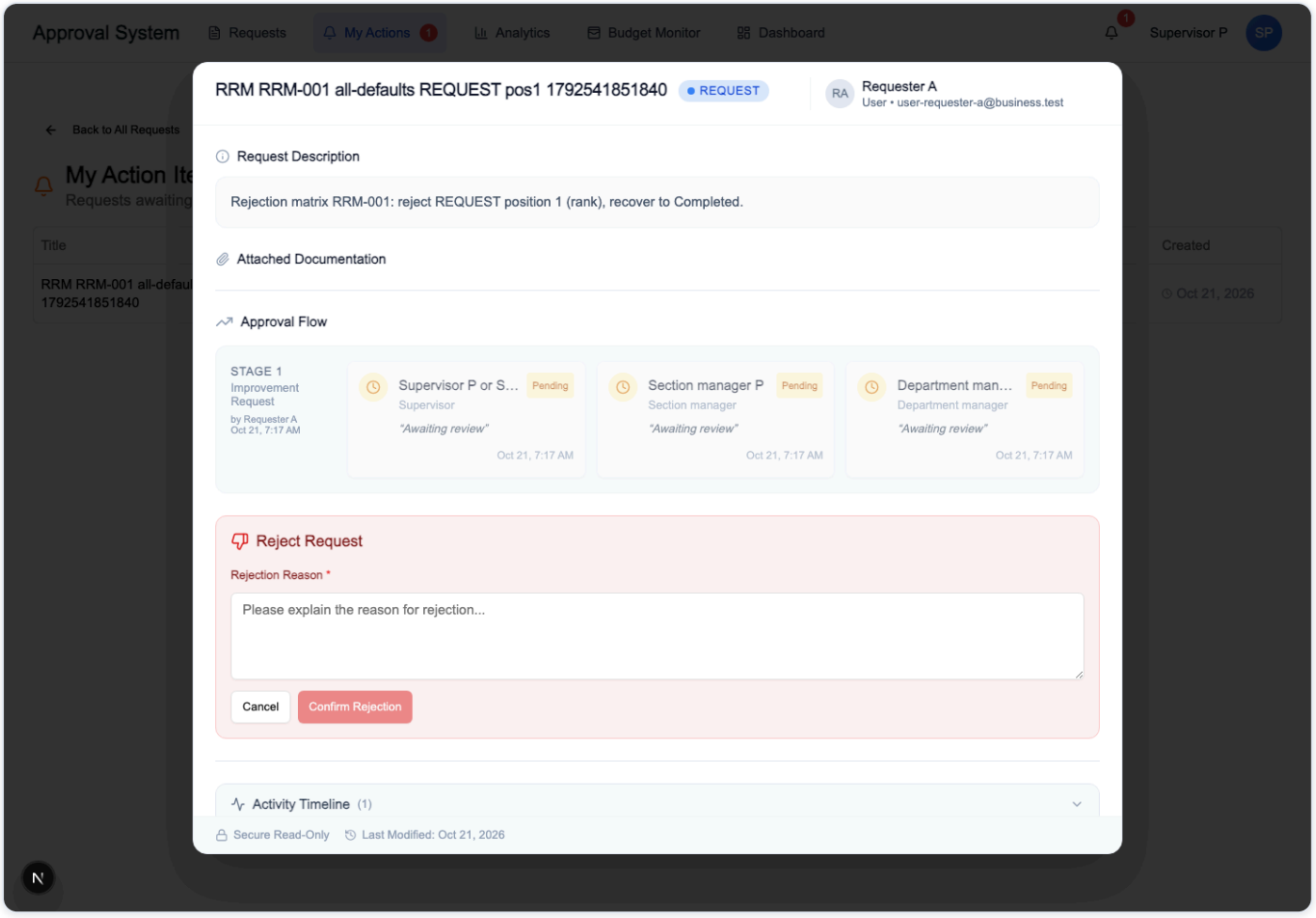


Figure 2. The approver's view: the whole approval flow, who is still pending, and the Reject box (opened here).

Approvers find the request under **My Actions** when it is their turn. They see the whole chain (here Supervisor → Section manager → Department manager) and can approve or reject. **Confirm Rejection stays disabled until a reason is typed.** Nobody can decide out of turn, and nobody approves their own request: if the requester is a Supervisor, the Supervisor step is skipped.

1.3 If an approver rejects the request

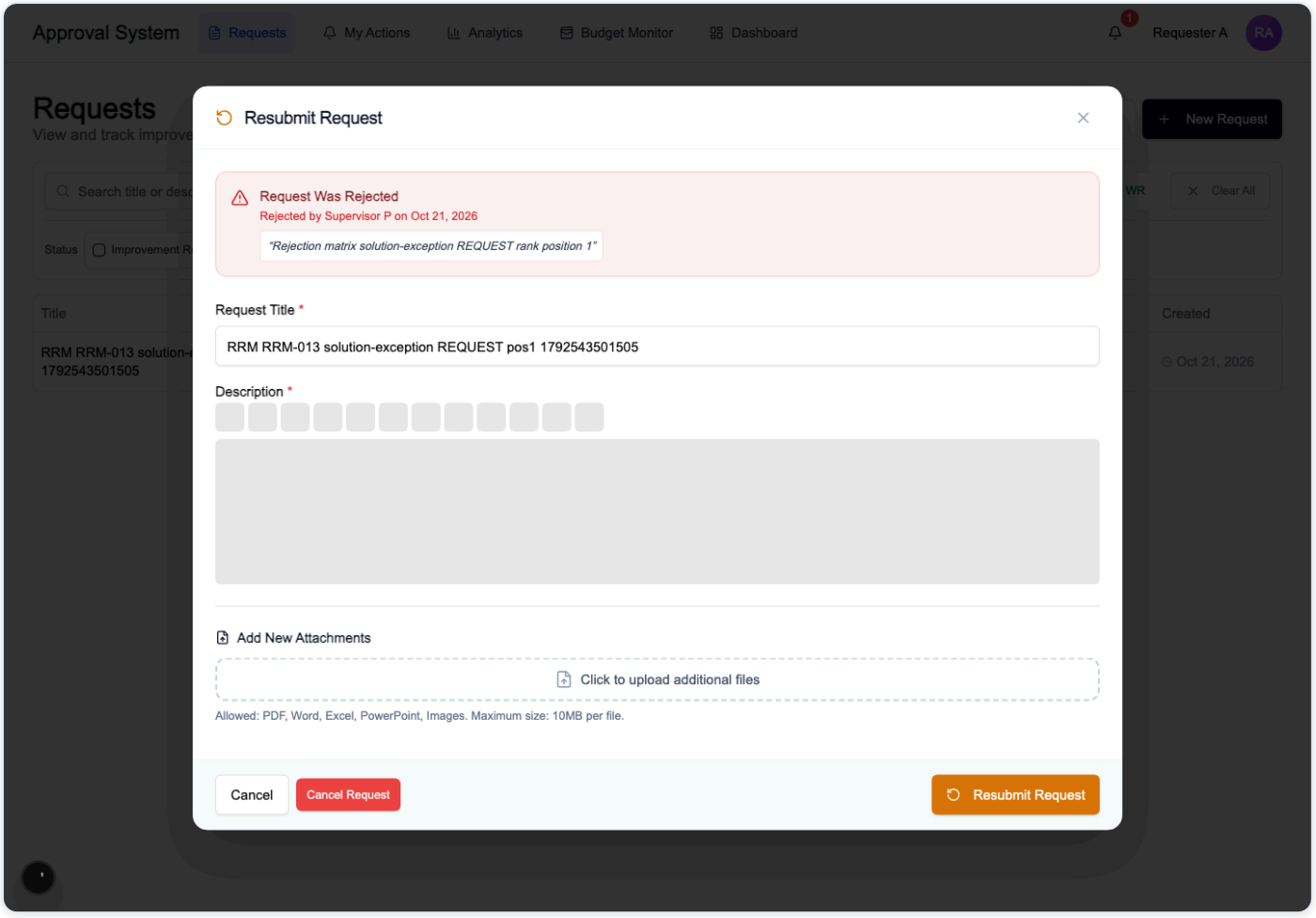


Figure 3. The requester sees who rejected the request, when and why, and can edit and resubmit it.

The requester sees **Request Was Rejected** with the name of the approver, the date and the reason. They fix the request and press **Resubmit Request**. The approvals start again from step 1; earlier approvals are not carried over.

Passing tests for Stage 1

Test	What it proves	Last passed
T01	A requester signs in and creates a request.	2026-10-06
T01	A Supervisor signs in and creates a request.	2026-10-06
T01	A deactivated user cannot sign in.	2026-10-06
T02A	Request approvals happen in order. Someone out of turn, or outside the approver group, cannot decide.	2026-10-06
T02B	When the requester is a Supervisor, the Supervisor step is skipped, so nobody approves their own request.	2026-10-06

Test	What it proves	Last passed
T14A	Rejecting a Request needs a reason; the requester resubmits; a non-permitted person is denied; replaying a stale rejection after the resubmit is denied.	2026-10-06

Rejection and recovery cases that reject at the Request stage (12, all passed 2026-10-21):

Case	Route setup	Rejected by	Chain	Result
RRM-001	All default chains	Supervisor (step 1)	Default (by position)	✅ Completed
RRM-002	All default chains	Section manager (step 2)	Default (by position)	✅ Completed
RRM-003	All default chains	Department manager (step 3)	Default (by position)	✅ Completed
RRM-013	Solution exception chain	Supervisor (step 1)	Default (by position)	✅ Completed
RRM-014	Solution exception chain	Section manager (step 2)	Default (by position)	✅ Completed
RRM-015	Solution exception chain	Department manager (step 3)	Default (by position)	✅ Completed
RRM-024	Final exception chain	Supervisor (step 1)	Default (by position)	✅ Completed
RRM-025	Final exception chain	Section manager (step 2)	Default (by position)	✅ Completed
RRM-026	Final exception chain	Department manager (step 3)	Default (by position)	✅ Completed
RRM-035	Solution + Final exception chains	Supervisor (step 1)	Default (by position)	✅ Completed
RRM-036	Solution + Final exception chains	Section manager (step 2)	Default (by position)	✅ Completed
RRM-037	Solution + Final exception chains	Department manager (step 3)	Default (by position)	✅ Completed

Stage 2: the Solution

When the last Request approver says yes, the request goes to Engineering.

2.1 Engineering submits a solution

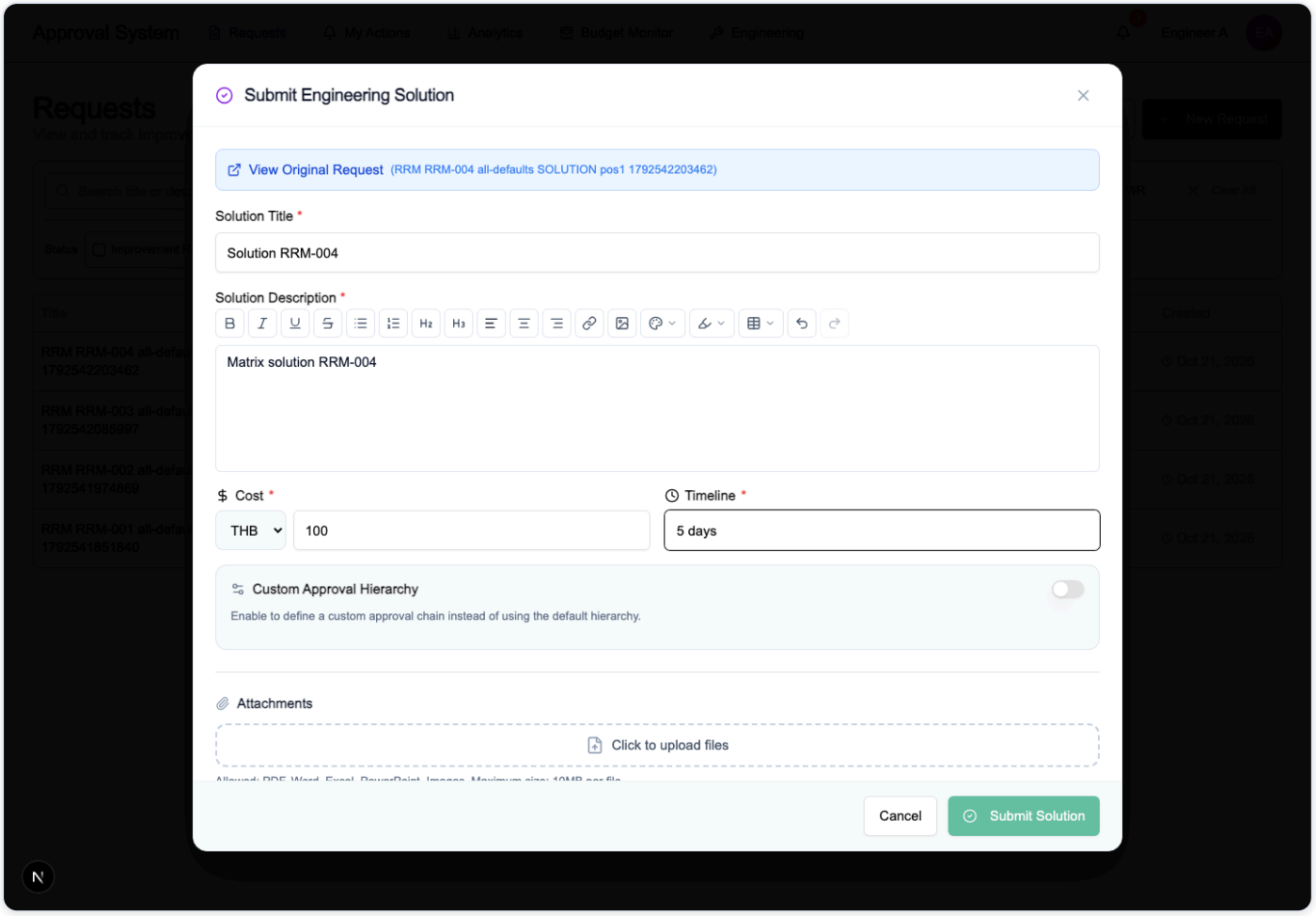


Figure 4. Submit Engineering Solution: title, description, cost, timeline and an optional custom approval hierarchy.

An engineer submits an **Engineering Solution**: a title, a description, the cost and the timeline. Staff of the requester's section, people from a sibling section and inactive users cannot submit one, and nothing is half-saved. The engineer can also define a custom approval chain for the solution (see "Custom approval chains").

2.2 The solution is approved

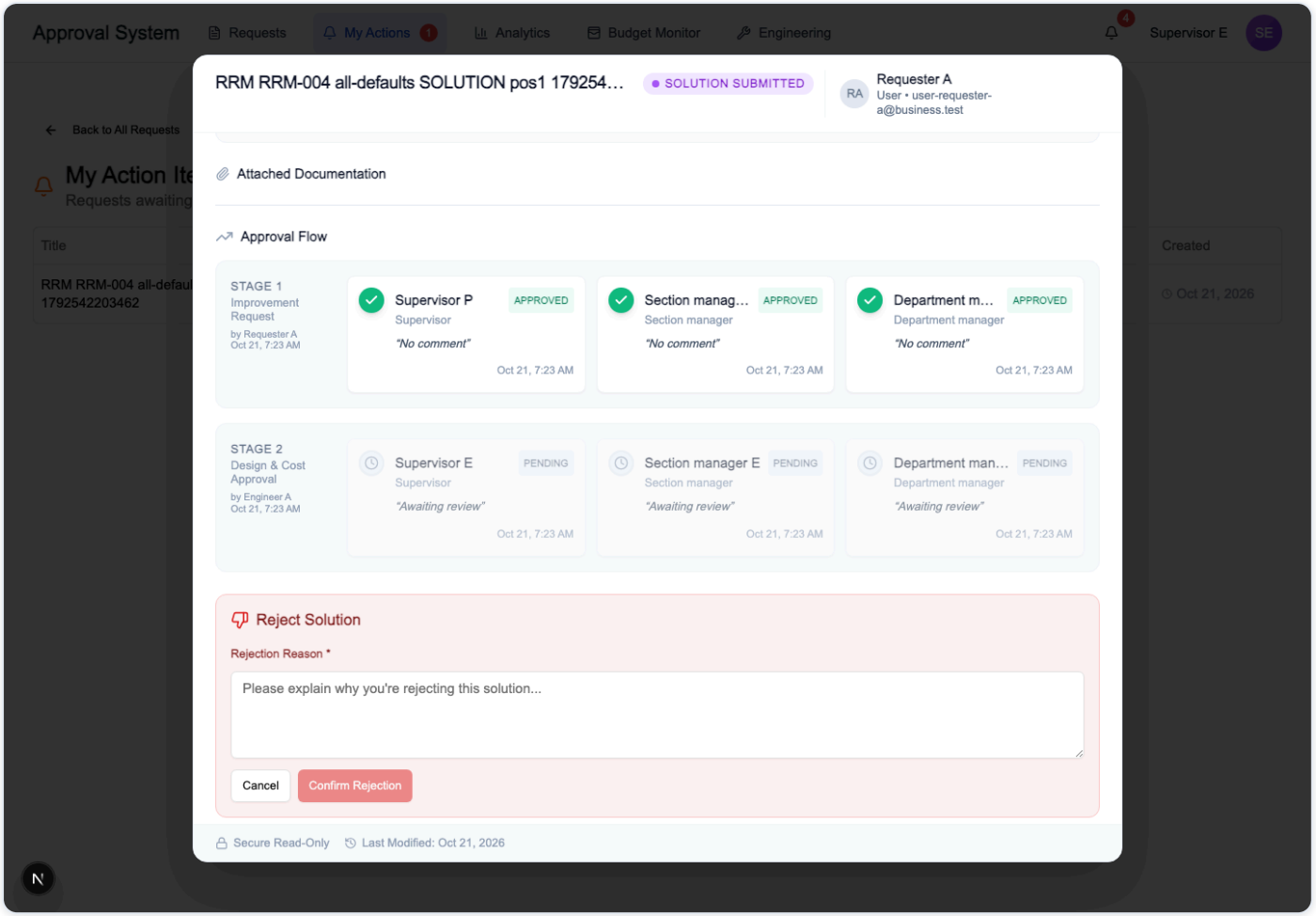


Figure 5. Stage 1 is complete (all approved); Stage 2, Design & Cost Approval, has three pending steps. The approver can approve or reject.

The solution goes through its own chain: here Engineering's Supervisor → Engineering's Section manager → Department manager. The approver sees the earlier Request approvals and the pending Solution steps, and approves or rejects (a reason is required).

2.3 If the solution is rejected

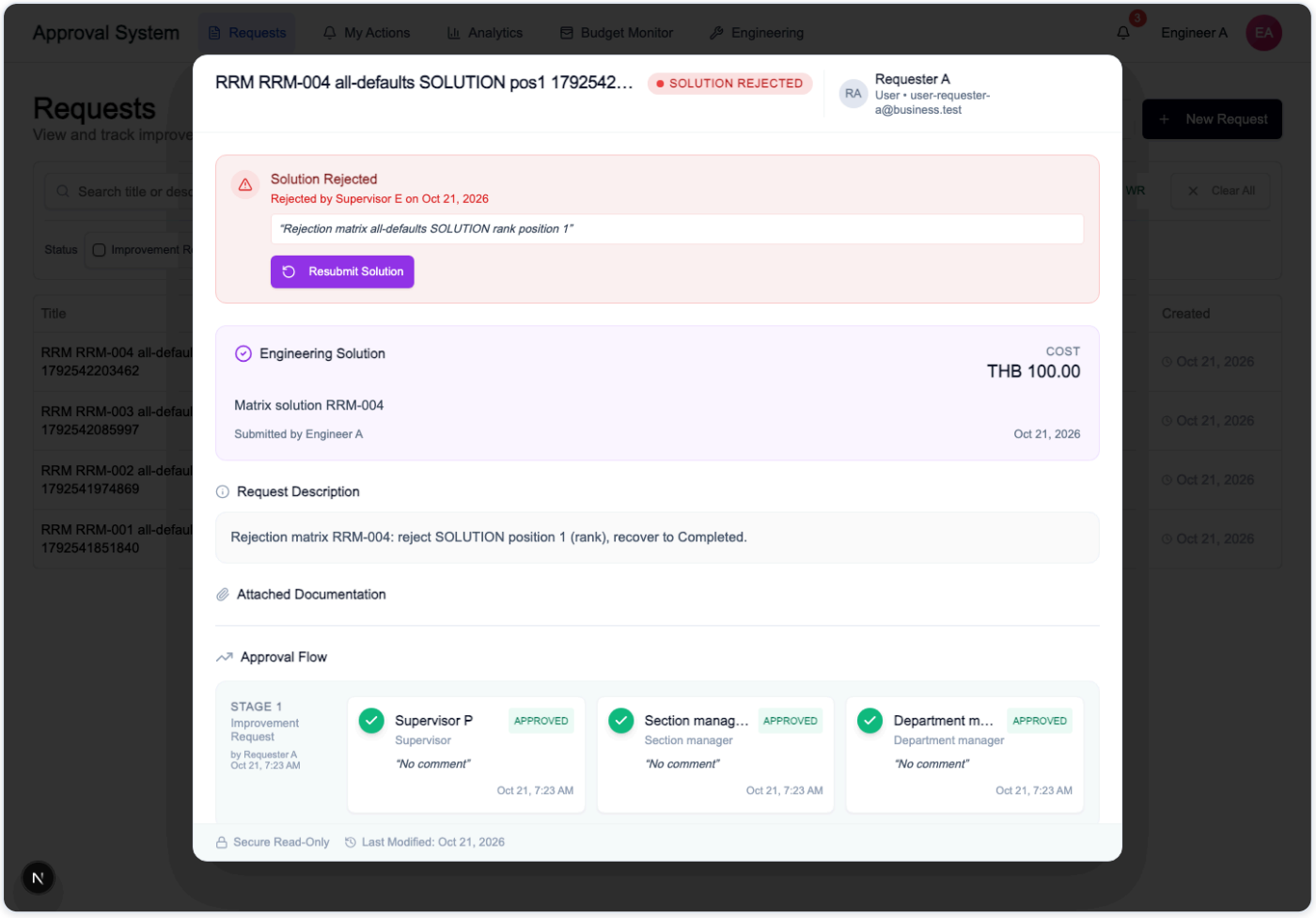


Figure 6. Solution Rejected: Engineering sees who rejected, when and why, with a Resubmit Solution button.

Engineering sees **Solution Rejected** with the reason and a **Resubmit Solution** button. After they resubmit, the Solution approvals start again from step 1.

Passing tests for Stage 2

Test	What it proves	Last passed
T03A	Engineering submits a solution; the first Solution approval goes to Engineering's Supervisor.	2026-10-06
T03B	When the engineer is a Supervisor, that step is skipped and the Section manager decides first.	2026-10-06
T03C	Staff, people from a sibling section and inactive users cannot submit a solution, and nothing is half-saved.	2026-10-06
T14B	Rejecting a Solution needs a reason; Engineering resubmits; a non-permitted person is denied.	2026-10-06

Rejection and recovery cases that reject at the Solution stage (18, all passed 2026-10-21):

Case	Route setup	Rejected by	Chain	Result
RRM-004	All default chains	Supervisor (step 1)	Default (by position)	✅ Completed
RRM-005	All default chains	Section manager (step 2)	Default (by position)	✅ Completed
RRM-006	All default chains	Department manager (step 3)	Default (by position)	✅ Completed
RRM-009	All default chains	Named approver #1	Exception chain (named people)	✅ Completed
RRM-010	All default chains	Named approver #2	Exception chain (named people)	✅ Completed
RRM-016	Solution exception chain	Section manager (step 1)	Default (by position)	✅ Completed
RRM-017	Solution exception chain	Department manager (step 2)	Default (by position)	✅ Completed
RRM-020	Solution exception chain	Named approver #1	Exception chain (named people)	✅ Completed
RRM-021	Solution exception chain	Named approver #2	Exception chain (named people)	✅ Completed
RRM-027	Final exception chain	Supervisor (step 1)	Default (by position)	✅ Completed
RRM-028	Final exception chain	Section manager (step 2)	Default (by position)	✅ Completed
RRM-029	Final exception chain	Department manager (step 3)	Default (by position)	✅ Completed
RRM-031	Final exception chain	Named approver #1	Exception chain (named people)	✅ Completed
RRM-032	Final exception chain	Named approver #2	Exception chain (named people)	✅ Completed
RRM-038	Solution + Final exception chains	Section manager (step 1)	Default (by position)	✅ Completed
RRM-039	Solution + Final exception chains	Department manager (step 2)	Default (by position)	✅ Completed
RRM-041	Solution + Final exception chains	Named approver #1	Exception chain (named people)	✅ Completed

Case	Route setup	Rejected by	Chain	Result
RRM-042	Solution + Final exception chains	Named approver #2	Exception chain (named people)	✅ Completed

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Stage 3: Final approval

After the solution is approved, the request goes back to the requester ("Solution Approved & Sent to Requester"). The requester now starts **Final approval**.

3.1 Who can start Final approval

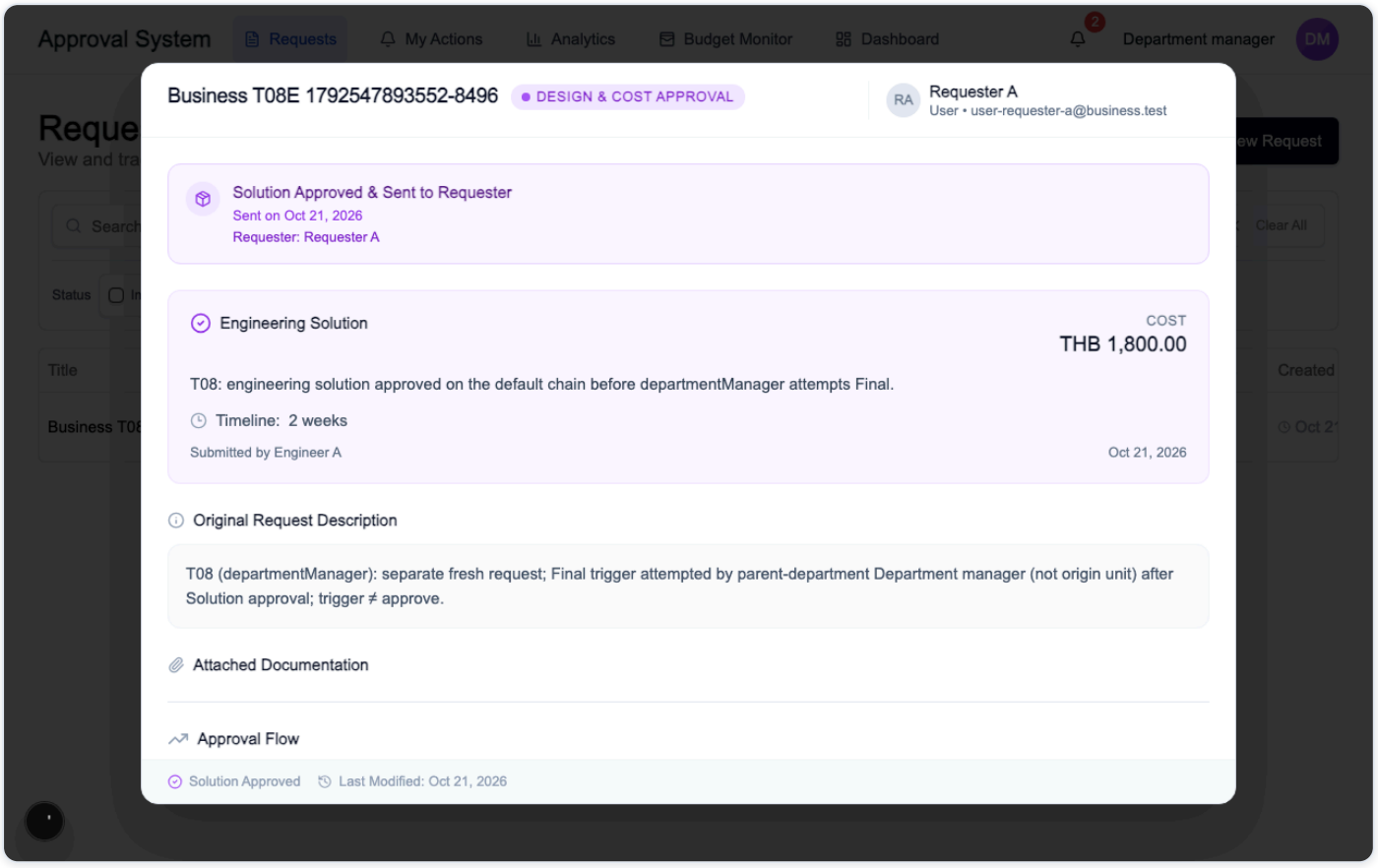


Figure 7. The Department manager of the parent department can see the request, but there is no way to start Final.

Only people in the requester's own unit can start Final approval. In this example the Department manager of the *parent* department opens the request and sees it, but has no control to start Final. The request stays as it is and no Final attempt is created. Starting Final also gives no right to approve it: starting and deciding are separate. Attempts by the solution author, by a sibling section, or by inactive or unassigned users are denied too, whether they come before the solution is approved or after the request is Completed.

3.2 The requester starts Final approval

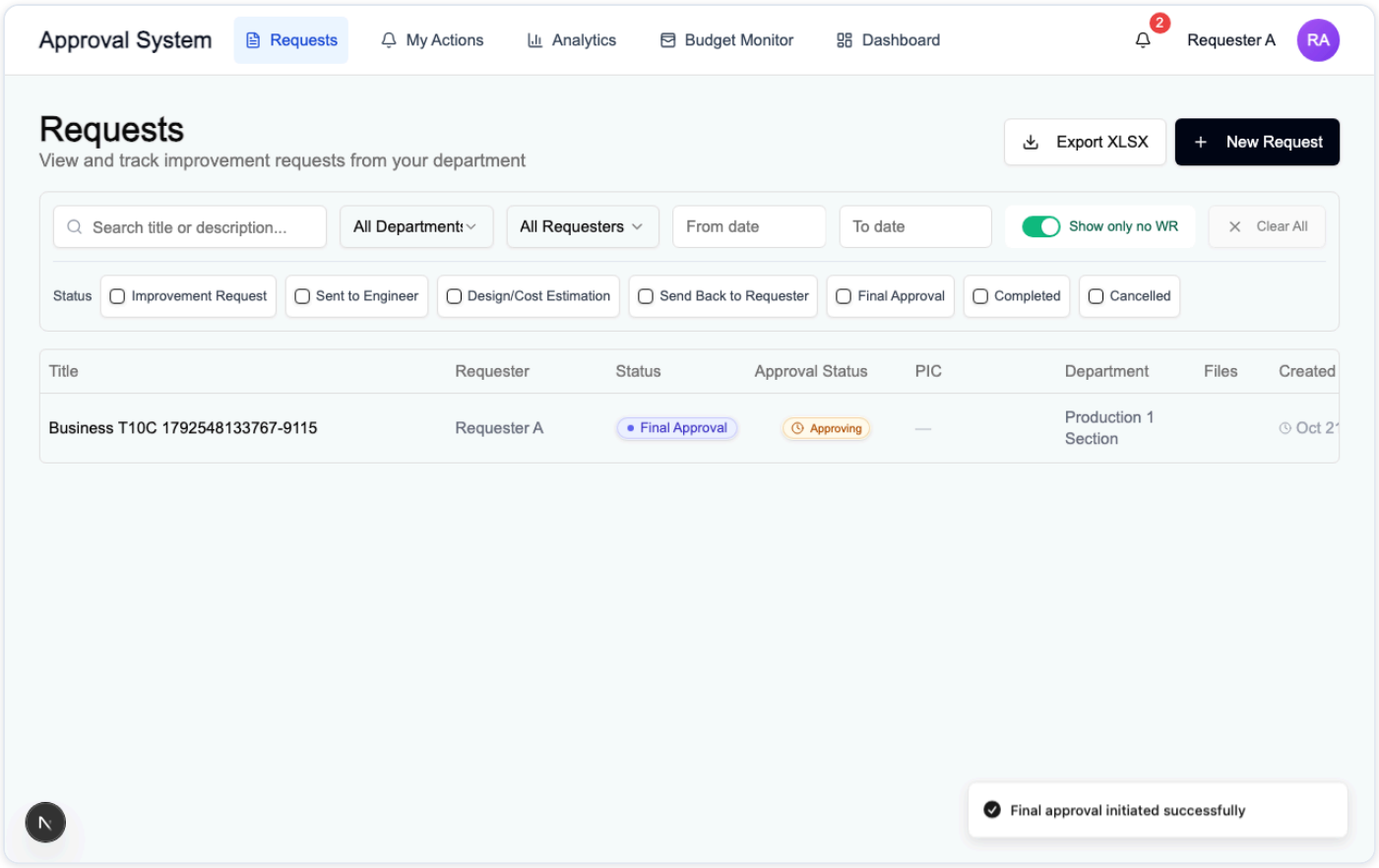


Figure 8. After the requester starts Final: the status becomes Final Approval · Approving, and the app confirms “Final approval initiated successfully”.

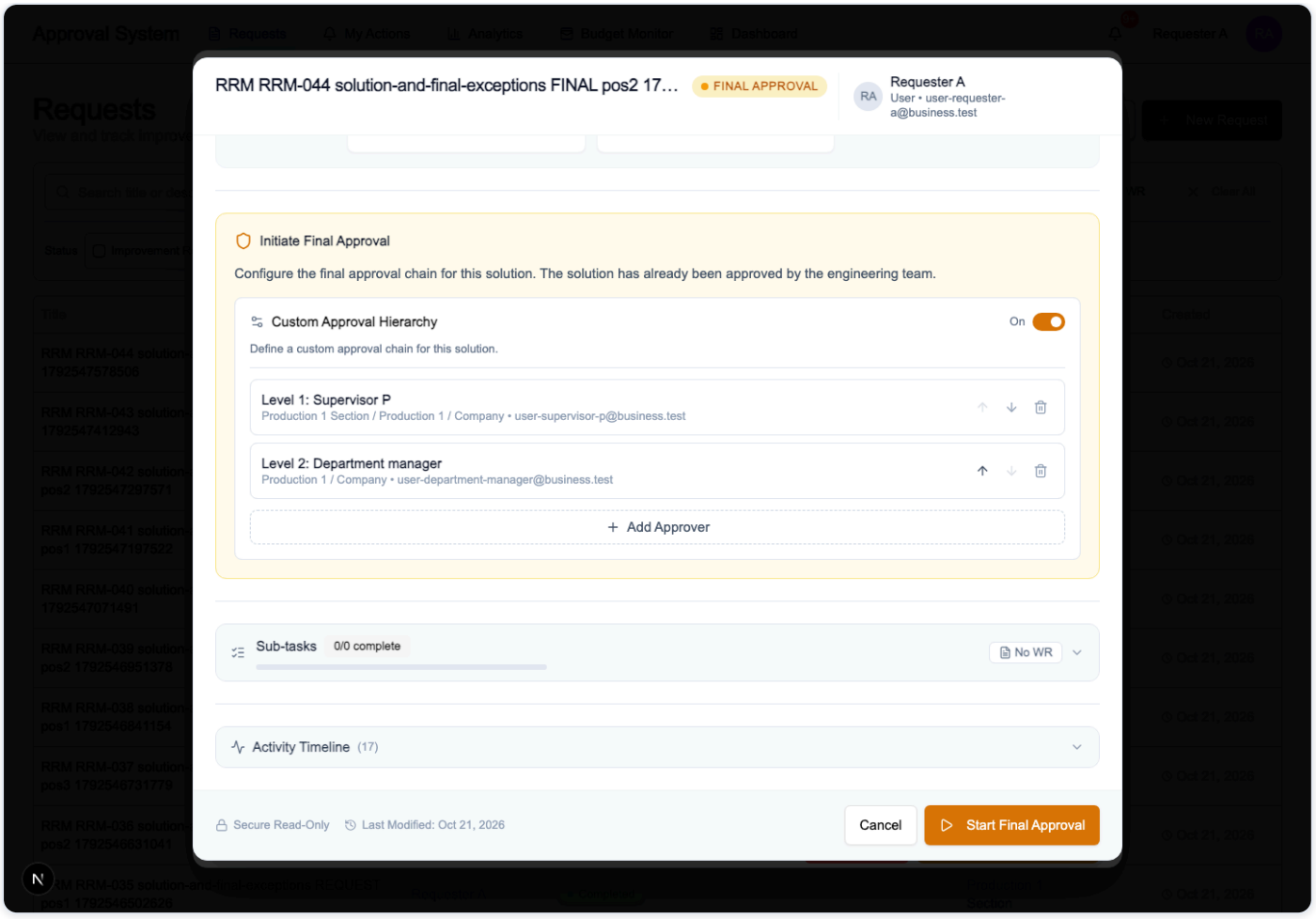


Figure 9. Starting Final: the requester can keep the default chain or define a custom one, level by level.

The status changes to **Final Approval · Approving** and the app confirms “Final approval initiated successfully”. When starting, the requester can keep the default chain or turn on **Custom Approval Hierarchy** and set Level 1, Level 2 and so on. Double-submitting the start produces exactly one transition.

3.3 Final approvers decide

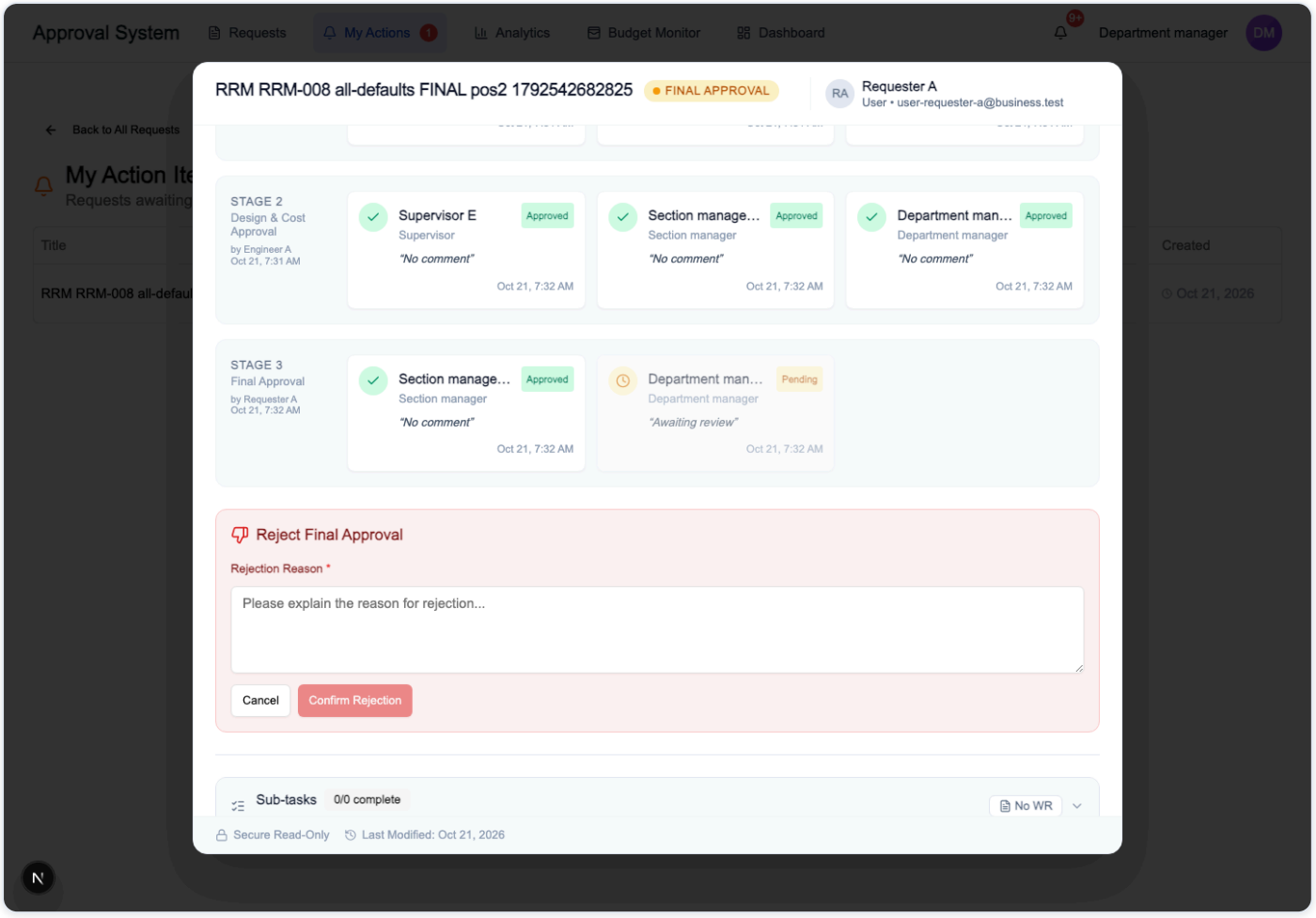


Figure 10. The Final approver's turn: Stage 3 shows the Section manager has approved; the Department manager is next and can approve or reject.

Final approvers decide in order. Here the Section manager has approved and the Department manager is next, who can approve or reject (a reason is required to reject).

3.4 If Final approval is rejected

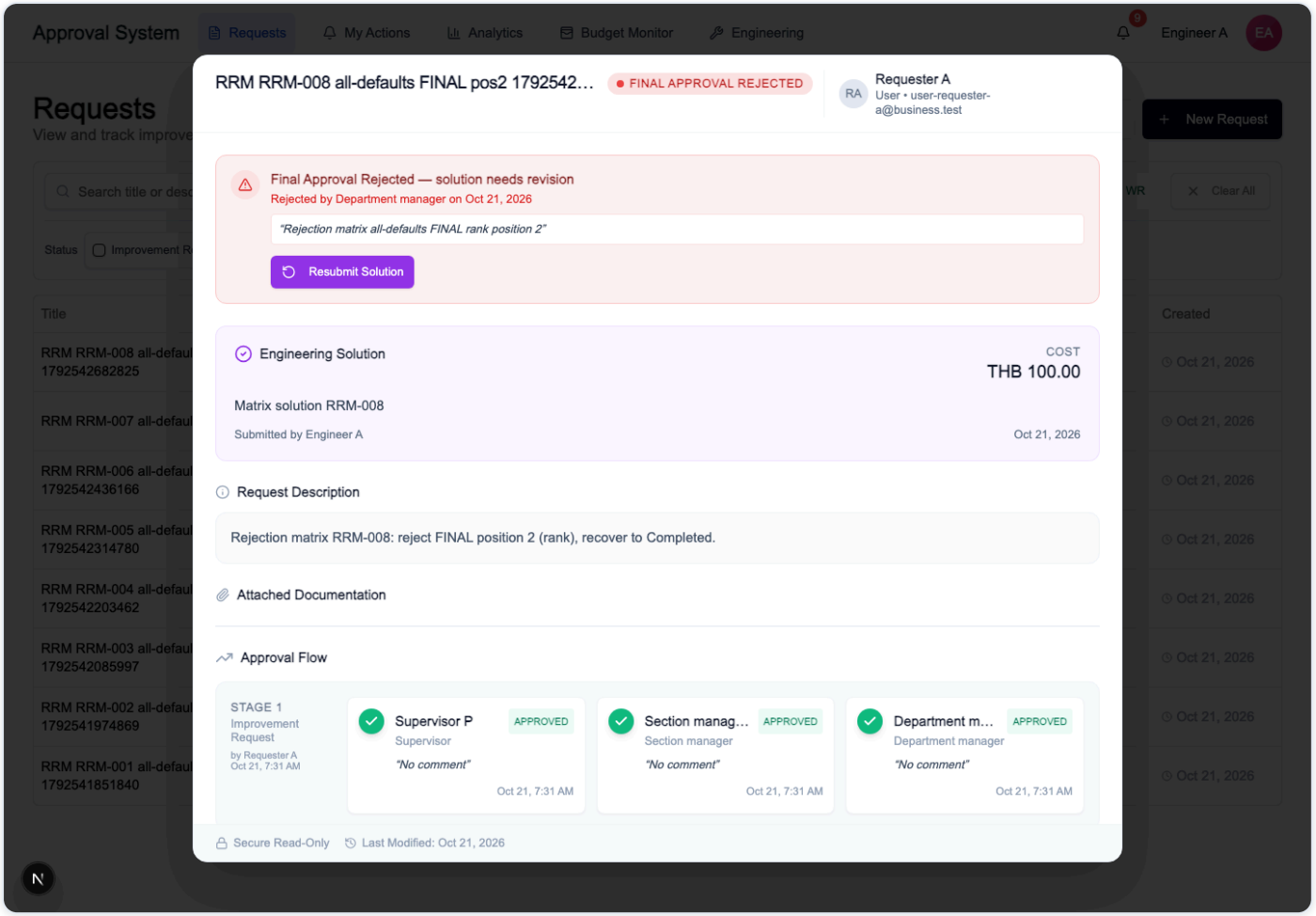


Figure 11. Final Approval Rejected: the solution goes back to Engineering for revision, with a Resubmit Solution button.

A rejection at Final sends the **solution** back to Engineering (“Final Approval Rejected — solution needs revision”). Engineering revises it and presses **Resubmit Solution**; the Solution approvals and then the Final approvals run again. The requester does not have to start over. Replaying an old “Confirm rejection” on the replaced attempt is denied and changes nothing.

Passing tests for Stage 3

Test	What it proves	Last passed
T08A	The requester starts Final. Starting Final gives no right to approve it; the default Final chain decides → Completed.	2026-10-06
T08B	A staff member of the requester’s unit starts Final; the default Final chain decides → Completed.	2026-10-06
T08C	A Supervisor of the requester’s unit starts Final; the default Final chain decides → Completed.	2026-10-06

Test	What it proves	Last passed
T08D	The Section manager of the requester's unit starts Final; the default Final chain decides → Completed.	2026-10-06
T08E	The Department manager of the parent department is denied the Final trigger (only the requester's own unit may start Final). The request stays "Sent Back to Requester" and no Final attempt is created.	2026-10-21 ✅ final code
T09A	Before the solution is approved, attempts to start Final by the solution author, a sibling section, an inactive user or an unassigned user are denied, with no attempt and no status change.	2026-10-06
T09B	After the request is Completed, attempts to start Final by the solution author, a sibling section, an inactive user or an unassigned user are denied, with no extra attempt.	2026-10-06
T10A	Final follows the original requester's branch when the requester starts it; the Final exception chain (Department manager only) decides → Completed.	2026-10-06
T10B	Same when a staff member starts Final: the original requester's branch is kept → Completed.	2026-10-06
T10C	The parent Department manager is denied; then the requester starts Final; the Final exception chain decides → Completed.	2026-10-21 ✅ final code
T14C	Rejecting at Final needs a reason; Engineering revises the solution; a non-permitted person is denied.	2026-10-06
T15A	Double-submitting the Final trigger produces exactly one transition.	2026-10-06

Rejection and recovery cases that reject at the Final stage (14, all passed 2026-10-21):

Case	Route setup	Rejected by	Chain	Result	Stale reject replayed
RRM-007	All default chains	Section manager (step 1)	Default (by position)	✅ Completed	Denied ✔
RRM-008	All default chains	Department manager (step 2)	Default (by position)	✅ Completed	Denied ✔
RRM-011	All default chains	Named approver #1	Exception chain (named people)	✅ Completed	Denied ✔
RRM-012	All default chains	Named approver #2	Exception chain (named people)	✅ Completed	Denied ✔
RRM-018	Solution exception chain	Section manager (step 1)	Default (by position)	✅ Completed	Denied ✔

Case	Route setup	Rejected by	Chain	Result	Stale reject replayed
RRM-019	Solution exception chain	Department manager (step 2)	Default (by position)	✅ Completed	Denied ✔
RRM-022	Solution exception chain	Named approver #1	Exception chain (named people)	✅ Completed	Denied ✔
RRM-023	Solution exception chain	Named approver #2	Exception chain (named people)	✅ Completed	Denied ✔
RRM-030	Final exception chain	Department manager (step 1)	Default (by position)	✅ Completed	Denied ✔
RRM-033	Final exception chain	Named approver #1	Exception chain (named people)	✅ Completed	Denied ✔
RRM-034	Final exception chain	Named approver #2	Exception chain (named people)	✅ Completed	Denied ✔
RRM-040	Solution + Final exception chains	Department manager (step 1)	Default (by position)	✅ Completed	Denied ✔
RRM-043	Solution + Final exception chains	Named approver #1	Exception chain (named people)	✅ Completed	Denied ✔
RRM-044	Solution + Final exception chains	Named approver #2	Exception chain (named people)	✅ Completed	Denied ✔

Completed

4.1 Everything approved

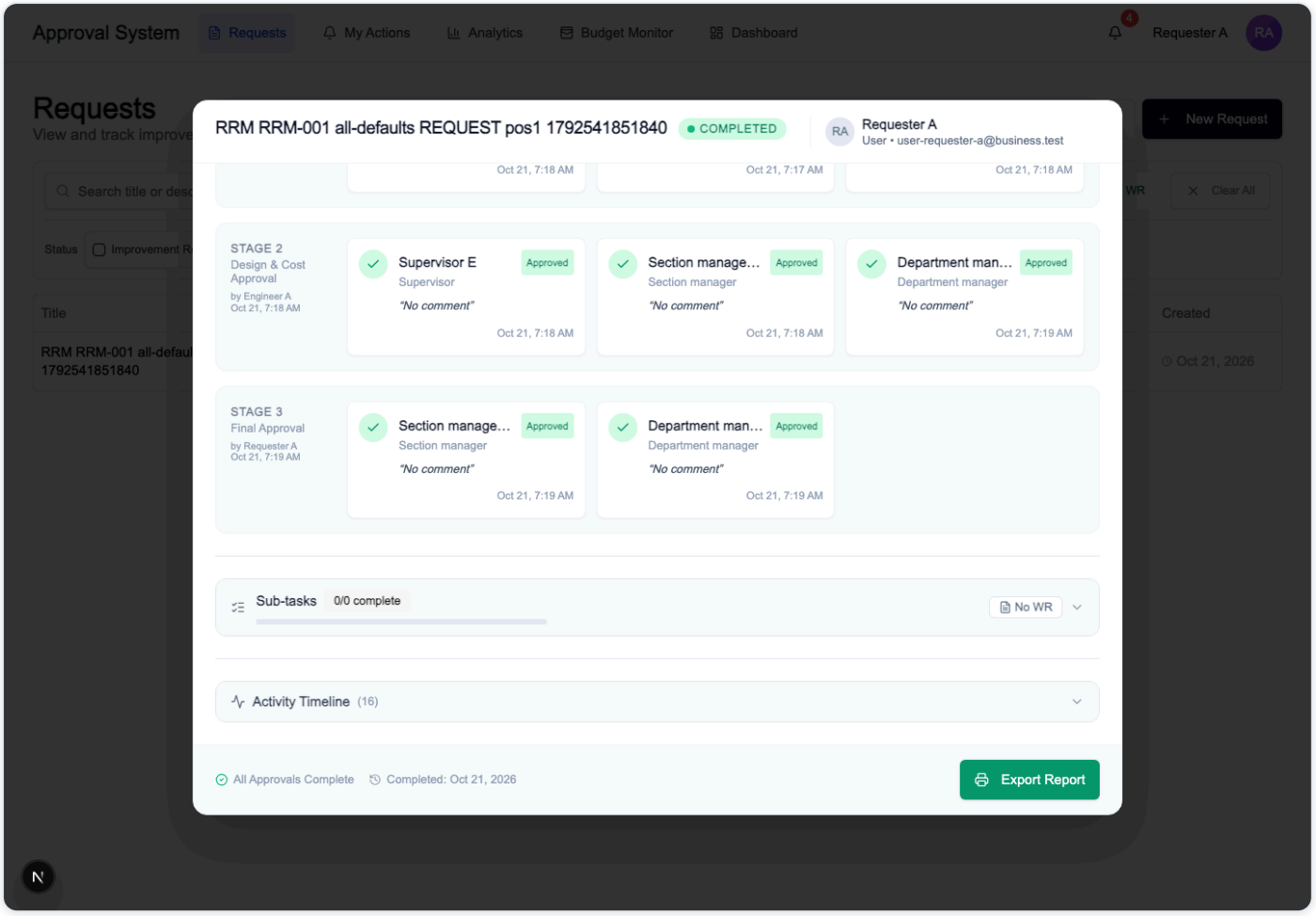


Figure 12. Completed: all approvals recorded, “All Approvals Complete”, and the report can be exported.

When the last approver says yes, the request is **Completed**. All approvals are recorded and the report can be exported. A completed request keeps its audit history and offers no further actions.

4.2 The history is kept

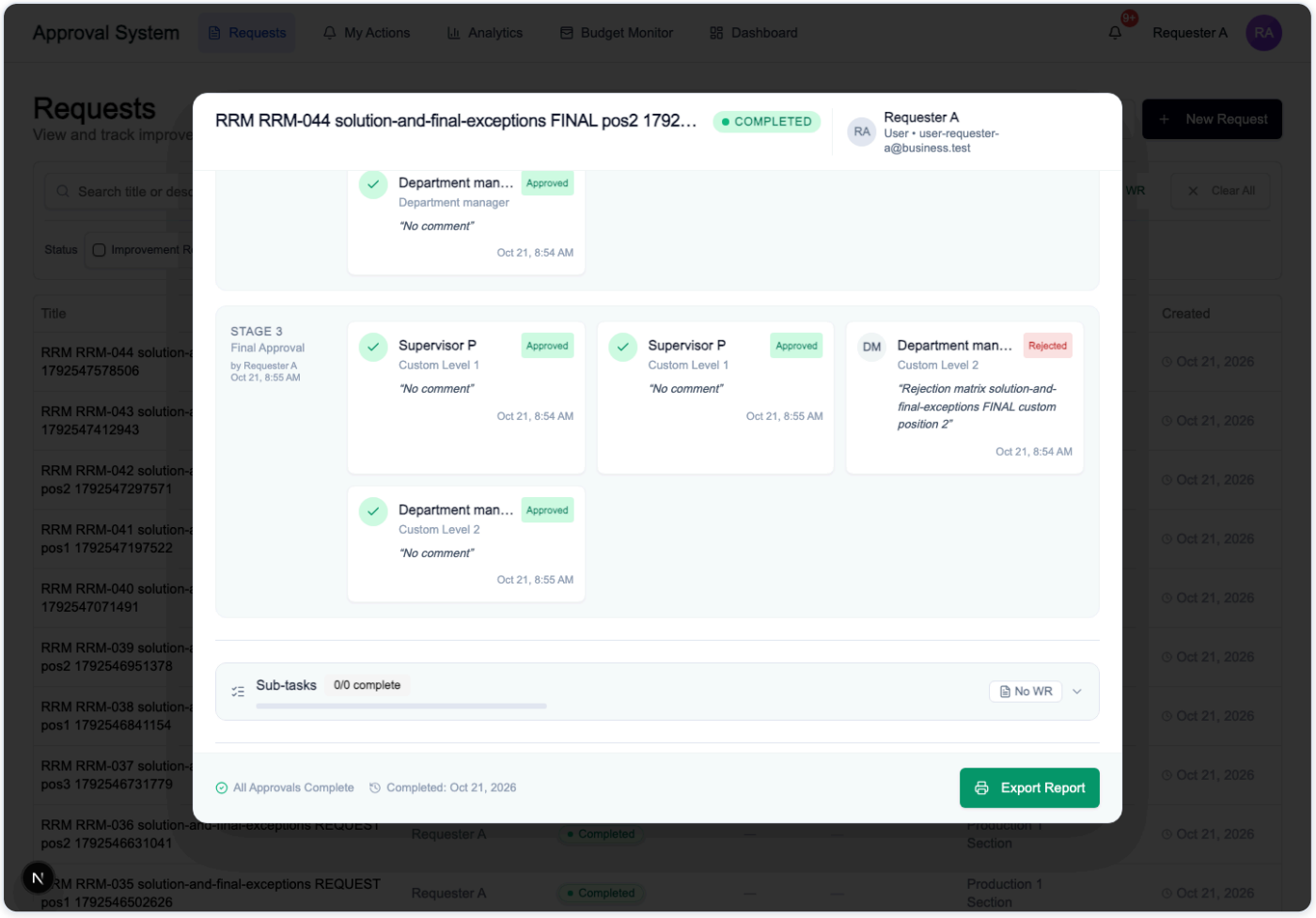


Figure 13. A rejected Final approval stays visible, with its reason, next to the approvals that followed after the fix (custom Level 1 and Level 2).

Rejections stay in the history. In this example a rejected approval in Stage 3 stays visible with its reason, next to the approvals that came after the fix.

Passing tests for the complete journey

Test	What it proves	Last passed
T04A	Full journey on the default chains, started by a staff member: request → solution → Final (started by the original requester) → Completed.	2026-10-06
T04B	Full journey where the requester and the engineer are Supervisors: their own steps are skipped; ends Completed.	2026-10-06
T20A	A completed request keeps its audit history and offers no further actions.	2026-10-06

Custom approval chains

Approval chains can be customized in two ways: an admin can set an **exception chain** for a unit and phase, and a person can define a **custom hierarchy** for one request when submitting a solution or starting Final (see the pictures in 2.1 and 3.2). Custom chains behave like the default ones, including rejection and recovery. The 44 rejection cases above run on four route setups: all default chains, a Solution exception, a Final exception, and both exceptions.

Pending requests keep the chain they were created with when the setup changes later; new requests follow the current setup. Invalid setups are refused before anything is saved.

Passing tests for custom chains

Test	What it proves	Last passed
T05	Custom Solution chain: only the replacement approvers decide; Request and Final keep the default chains.	2026-10-06
T06	Custom Final chain: the Department manager alone decides Final; Request and Solution keep the default chains.	2026-10-06
T07A	Custom Solution and Final chains together, started by a staff member, through to Completed.	2026-10-06
T07B	Custom Solution and Final chains started by Supervisors: they never approve their own work; ends Completed.	2026-10-06
T17A	Changing the Request route while a request is pending: the pending attempt keeps its captured chain; new requests follow the current setup.	2026-10-06
T17B1	A pending Solution attempt keeps its captured chain when the Solution exception changes.	2026-10-06
T17B2	After a Solution exception is removed, a new Solution attempt uses the current default.	2026-10-06
T17C1	A pending Final attempt keeps its captured chain when the Final exception changes.	2026-10-06
T17C2	After a Final exception is removed, a new Final attempt uses the current default.	2026-10-06
T19A	Invalid route drafts (duplicates, rank order, limit) are refused before anything is written.	2026-10-06
T19B	A duplicate unit/phase exception is refused and unauthorized publishing is denied; the last valid setup stays usable.	2026-10-06

Other protections

Safeguards around the main flow: who can see a request, who is notified, what happens when two people act at once, when something fails, when an approver is deactivated, and when a request is cancelled.

Test	What it proves	Last passed
T11A	With sharing ON while the request awaits Engineering, people in scope can see it but get no action rights; sibling and inactive users cannot see it.	2026-10-06
T12	In-app notifications reach the right people when a request is created, when the first approval is needed, when the solution is ready, when Final is needed and when it completes. A viewer gets none.	2026-10-06
T15B	Two eligible approvers race for the same step: one wins, the other gets a conflict.	2026-10-06
T15C	Approve and reject race: one decision wins, the other is denied.	2026-10-06
T15D	Retrying a stale decision form after someone else already acted is denied.	2026-10-06
T16A	A mail failure after the approval is saved does not undo the approval.	2026-10-06
T16B	A failed decision writes no success state and sends no success notification.	2026-10-06
T18A	Deactivating a saved approver: no session, no silent substitution; the remaining candidate decides.	2026-10-06
T18B	A stranded attempt: an unauthorized restart is denied; an admin restart (with a reason) rebuilds the chain from the current setup.	2026-10-06
T18C1	A required position that is vacant from the start is skipped and recorded, never silently substituted.	2026-10-06
T18C2	If every step is vacant at creation, the request completes without approvals (recorded product behavior).	2026-10-06
T20B	The requester can cancel where policy allows; a cancelled request is locked down.	2026-10-06

Admin area security

Who can reach the admin area and what happens when someone who should not tries. All 11 tests passed.

What it proves	Last passed
Anonymous visitors: admin pages redirect to sign-in, and all ten admin actions redirect without leaking anything.	2026-09-21
Unauthorized people: all ten admin actions are denied and nothing changes.	2026-09-21
A general-department user denied the admin area lands on Requests and keeps their session.	2026-09-21
An Engineering user denied the admin area is sent to the Engineering page and keeps their session.	2026-09-21
Engineering membership without the Engineering role still lands on Requests, not on the admin area.	2026-09-21
Retention admin APIs: unauthorized people are denied; authorized controls get past authentication.	2026-09-21
A standby admin can sign in while active (used for the later deactivation check).	2026-09-21
An authorized admin: pages render, all ten guarded actions resolve, and read controls return real data.	2026-09-21
An authorized admin: the real controls change test data, and the six data-changing actions are captured.	2026-09-21
An admin deactivated through the real UI is denied on all ten admin actions.	2026-09-21
Final read-back: replaying the denied actions changed nothing.	2026-09-21